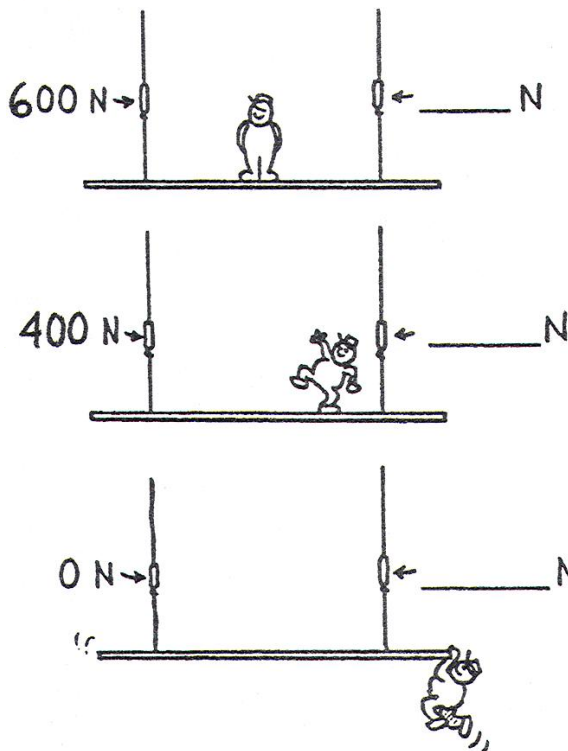
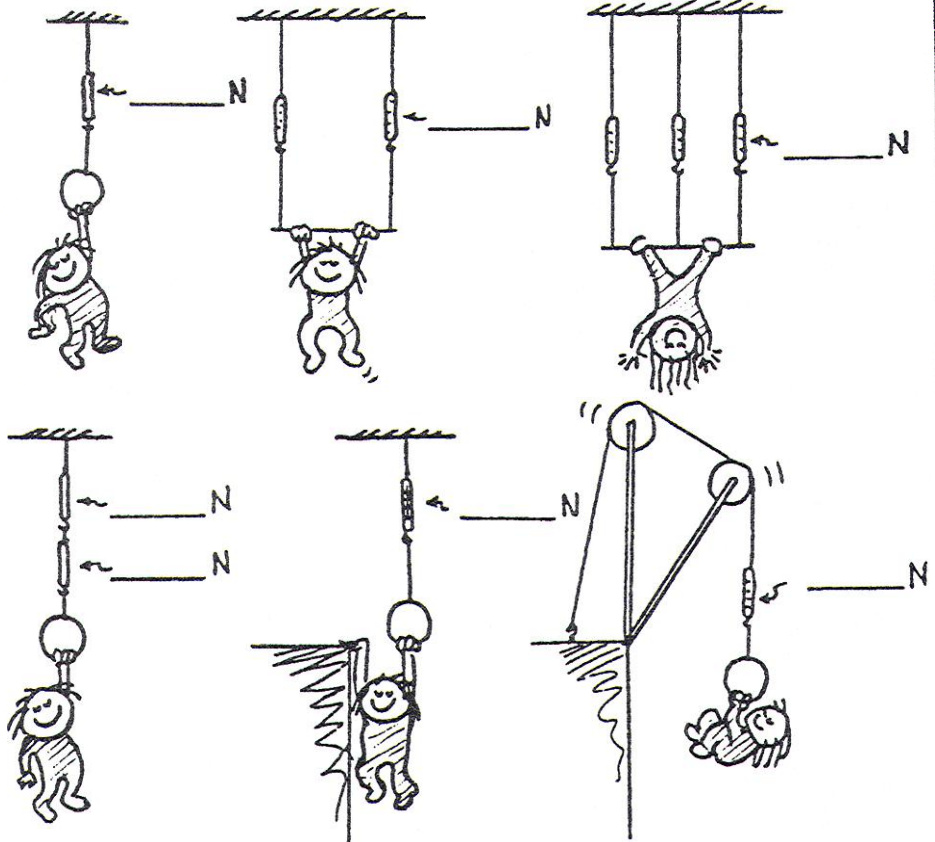


CONCEPTUAL *Physics* PRACTICE PAGE

Chapter 2 Newton's First Law of Motion—Inertia Static Equilibrium

1. Little Nellie Newton wishes to be a gymnast and hangs from a variety of positions as shown. Since she is not accelerating, the net force on her is zero. That is, $\Sigma \mathbf{F} = 0$. This means the upward pull of the rope(s) equals the downward pull of gravity. She weighs 300 N. Show the scale reading(s) for each case.



2. When Burl the painter stands in the exact middle of his scaffold, the left scale reads 600 N. Fill in the reading on the right scale. The total weight of Burl and staging must be _____ N.

3. Burl stands farther from the left. Fill in the reading on the right scale.

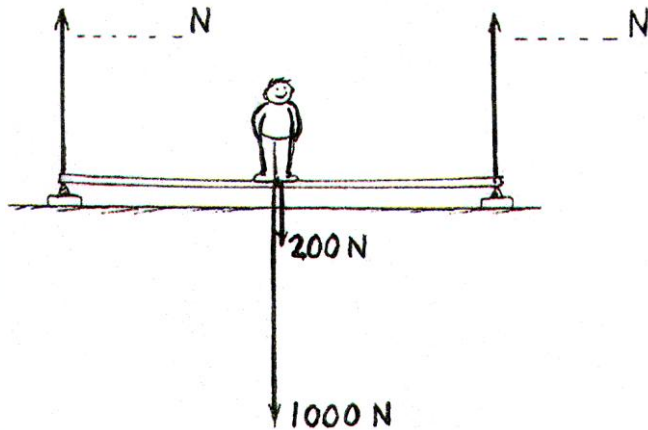
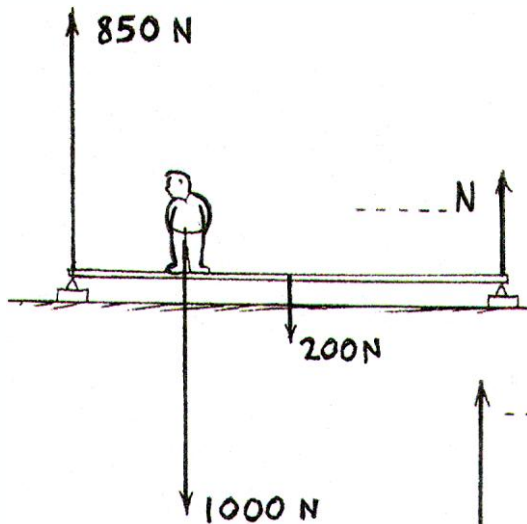
4. In a silly mood, Burl dangles from the right end. Fill in the reading on the right scale.

Hewitt
Draw it!

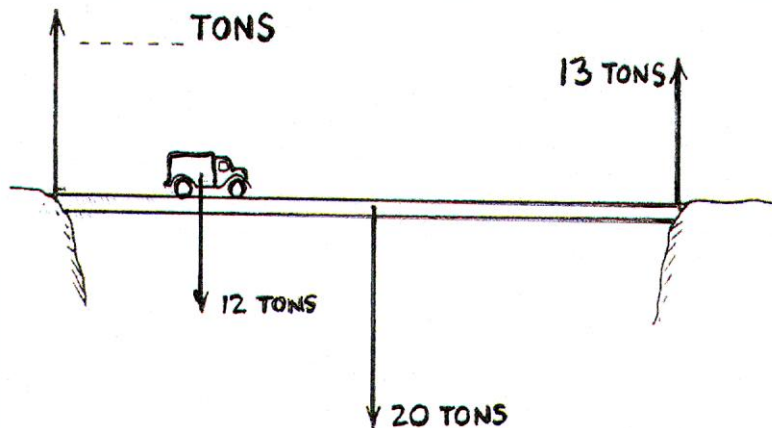
Chapter 2 Newton's First Law of Motion—Inertia

The Equilibrium Rule: $\Sigma F = 0$

1. Manuel weighs 1000 N and stands in the middle of a board that weighs 200 N. The ends of the board rest on bathroom scales. (We can assume the weight of the board acts at its center.) Fill in the correct weight reading on each scale.

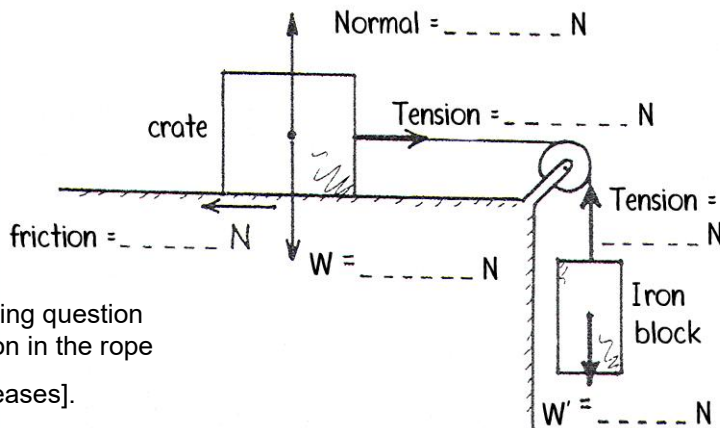


2. When Manuel moves to the left as shown, the scale closest to him reads 850 N. Fill in the weight for the far scale.



3. A 12-ton truck is one-quarter the way across a bridge that weighs 20 tons. A 13-ton force supports the right side of the bridge as shown. How much support force is on the left side?

4. A 1000-N crate resting on a surface is connected to a 500-N block through a frictionless pulley as shown. Friction between the crate and surface is enough to keep the system at rest. The arrows show the forces that act on the crate and the block. Fill in the magnitude of each force.



Circle the correct answers.

5. If the crate and block in the preceding question move at constant speed, the tension in the rope
[is the same] [increases] [decreases].

The sliding system is then in [static equilibrium] [dynamic equilibrium].

Hewitt
Draw it!

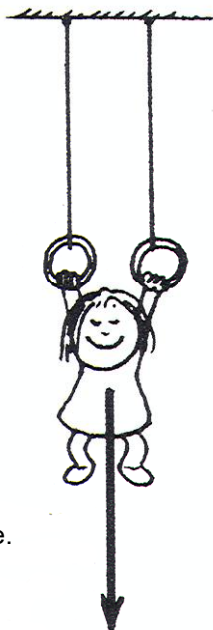
CONCEPTUAL *Physics* PRACTICE PAGE

Chapter 2 Newton's First Law of Motion—Inertia Vectors and Equilibrium



Nellie Newton dangles from a vertical rope in equilibrium: $\Sigma \mathbf{F} = 0$. The tension in the rope (upward vector) has the same magnitude as the downward pull of gravity (downward vector).

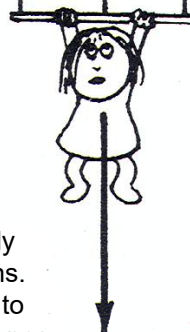
1. Nellie is supported by two vertical ropes. Draw tension vectors to scale along the direction of each rope.



2. This time the vertical ropes have different lengths. Draw tension vectors to scale for each of the two ropes.



3. Nellie is supported by three vertical ropes that are equally taut but have different lengths. Again, draw tension vectors to scale for each of the three ropes.



4. We see that tension in a rope is [dependent on] [independent of] the length of the rope. So the length of a vector representing rope tension is [dependent on] [independent of] the length of the rope.



Rope tension does depend on the angle the rope makes with the vertical, as Practice Pages for Chapter 5 will show!

Hewitt
Draw it!

I'd rather hang out with friends
who have reasonable doubts than
ones who are absolutely certain
about everything.



Hewitt
Drew it!